PRE-ALGEBRA (7TH) · UNIT 11: RIGHT TRIANGLES IN ALGEBRA

Zip Line Engineer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students engineer a zip line. Square roots and the Pythagorean theorem give cable and wire lengths, the distance and midpoint formulas locate stations on a site map, and trig ratios set a safe descent angle.

The 3° to 8° safety rule makes tangent meaningful.

Standards Addressed

- 8.EE.A.2 Square roots
- 8.G.B.6–8 Pythagorean theorem, converse, and distance
- G-SRT.C.6–8 Trig ratios and special right triangles (Honors/CP1 preview)
- 7.G.A.1 Similar figures and proportions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a zip line video. Part 1 square roots and cable lengths.
Day 2	Part 2 site map, distance, and midpoint.
Day 3	Part 3 angles and the engineering report.

Materials

- Student handout and Site Map Cards
- Scientific calculators
- Rulers and protractors
- Grid paper

How the Three Versions Differ

Level	What changes
CP2	Perfect squares, Pythagorean triples, distances on a grid, and a right-triangle check.
CP1	Large cable lengths, estimation, distance and midpoint formulas, tangent angle, and similar triangles.
Honors	Distance range from an angle constraint, two landing options, special right triangle supports, and a justified recommendation.

Watch For (Common Misconceptions)

- Adding the legs instead of their squares.
- Using the hypotenuse as a leg.
- Calculator in radian mode.

Extension Ideas

- Research real zip line angles.
- Build a string model with a protractor.
- Find the speed at the bottom using a simple physics formula.

Answer Key – CP2

Part 1: Square Roots

#	Task	Answer
1	Find √144 and √81.	12 and 9
2	Between which two whole numbers is √50? Which is it closer to?	7 and 8; closer to 7 (√49 = 7)
3	A square platform has area 64 sq ft. How long is each side?	8 ft
4	Estimate √200 to the nearest whole number.	14 ($14^2 = 196$)

Part 2: Cable Length

#	Task	Answer
5	A practice zip line drops 30 ft over a ground distance of 40 ft. How long is the cable?	$c^2 = 30^2 + 40^2 = 2{,}500 \rightarrow c = 50$ ft
6	A support wire runs from the top of a 12 ft post to a stake 5 ft away. How long is the wire?	13 ft
7	A 25 ft ladder reaches a tower 24 ft up. How far is the base of the ladder from the tower?	$\sqrt{(625 - 576)} = 7$ ft
8	Is a triangle with sides 9, 12, and 15 a right triangle? Show your check.	$81 + 144 = 225 = 15^2$, yes

Part 3: Site Map

#	Task	Answer
9	Plot the Tower (1, 2) and Lodge (1, 8). How far apart are they in units and in feet?	6 units = 300 ft
10	Use the Pythagorean theorem to find the distance from the Tower (1, 2) to the Landing (9, 8).	Legs 8 and 6 → 10 units = 500 ft
11	Draw your zip line on the map and write one safety rule for riders.	Answers vary.

Reflection

#	Prompt	Look for
R1	Why does the Pythagorean theorem only work for right triangles?	It relies on the right angle; other triangles do not have $a^2 + b^2 = c^2$.
R2	How did the grid help you find a slanted distance?	The horizontal and vertical changes form the legs of a right triangle.

Answer Key – CP1

Part 1: Cable Design

#	Task	Answer
1	Find the length of the cable to the nearest foot.	$\sqrt{(48^2 + 500^2)} = \sqrt{252{,}304} \approx 502$ ft
2	Estimate $\sqrt{252{,}304}$ without a calculator using perfect squares.	$502^2 = 252{,}004$ and $503^2 = 253{,}009$, so about 502.3
3	The cable sags, so the camp buys 5% extra. How much cable should they buy?	$502.3 \times 1.05 \approx 527$ ft
4	A support wire makes a 45°-45°-90° triangle with a 20 ft post. How long is the wire?	$20\sqrt{2} \approx 28.3$ ft

Part 2: Site Map

#	Task	Answer
5	Find the distance from Tower (1, 2) to Landing (9, 8) using the distance formula, in feet.	$\sqrt{(64 + 36)} = 10$ units = 500 ft
6	Find the midpoint of the zip line. Which station is closest to it?	(5, 5): First Aid is exactly at the midpoint
7	Find the distance from Landing (9, 8) to Boat Dock (10, 2) in feet (nearest foot).	$\sqrt{(1 + 36)} = \sqrt{37} \approx 6.08$ units ≈ 304 ft
8	Is the triangle Tower–Landing–Lodge a right triangle? Use the converse.	Tower–Lodge 6, Lodge–Landing 8, Tower–Landing 10; 36 + 64 = 100, yes

Part 3: Safe Angle

#	Task	Answer
9	Use tangent to find the angle of descent for a 48 ft drop over 500 ft.	$\tan\theta = \frac{48}{500} = 0.096 \rightarrow \theta \approx 5.5°$; safe
10	What is the greatest drop allowed over 500 ft for an 8° angle?	$500 \tan 8° \approx 70.3$ ft
11	At 4 ft tall, a camper casts a 6 ft shadow while the tower casts a 54 ft shadow. Use similar triangles to find the tower height.	$\frac{4}{6} = \frac{h}{54} \rightarrow h = 36$ ft

Reflection

#	Prompt	Look for
R1	How does the converse of the Pythagorean theorem help engineers?	It checks whether a corner is a right angle by testing the side lengths.

#	Prompt	Look for
R2	Why use tangent to find the descent angle?	The drop and horizontal distance are the opposite and adjacent sides.

Answer Key – Honors

Part 1: Design Constraints

#	Task	Answer
1	Find the range of horizontal distances that keep the angle safe.	48 ÷ tan 8° ≈ 341.5 ft to 48 ÷ tan 3° ≈ 915.9 ft
2	Which possible landing points on the Site Map (1 unit = 50 ft) from the Tower (1, 2) are within that range?	Landing 500 ft ✓; Boat Dock 450 ft ✓; Lodge 300 ft ×; First Aid √(16 + 9) = 5 units = 250 ft ×; Parking 150 ft ×.
3	For the Boat Dock landing, find the cable length and the descent angle.	Horizontal 450 ft; cable $\sqrt{(450^2 + 48^2)} \approx 452.6$ ft; $\theta = \tan^{-1}\left(\frac{48}{450}\right) \approx 6.1°$
4	Riders need at least 4 seconds of ride time at about 25 ft per second. Which landing meets this?	Need cable ≥ 100 ft; both do. Landing (≈ 502 ft) gives ≈ 20 s; Boat Dock ≈ 18 s.

Part 2: Support Structure

#	Task	Answer
5	Two support wires run from the top of a 36 ft tower to the ground, one at 60° and one at 45° with the ground. Find each wire length and how far each stake is from the tower base.	60°: wire $\frac{36}{\sin 60°} = 24\sqrt{3} \approx 41.6$ ft, stake $12\sqrt{3} \approx 20.8$ ft. 45°: wire $36\sqrt{2} \approx 50.9$ ft, stake 36 ft.
6	What is the distance between the two stakes if they are on opposite sides of the tower?	20.8 + 36 = 56.8 ft
7	A brace forms a 30°-60°-90° triangle with hypotenuse 10 ft. Find both legs exactly.	5 ft and $5\sqrt{3}$ ft
8	Prove the 45°-45°-90° ratio using the Pythagorean theorem.	Legs x, x: $x^2 + x^2 = c^2 \rightarrow c = x\sqrt{2}$

Part 3: Map Analysis

#	Task	Answer
9	Find the perimeter of the triangle Tower–Landing–Boat Dock in feet.	Tower–Landing 500, Landing–Dock $\sqrt{37}$ units ≈ 304 ft, Dock–Tower 450 ft → ≈ 1,254 ft
10	The first-aid station must be within 300 ft of the midpoint of any zip line built. For the Boat Dock zip line, check this.	Midpoint (5.5, 2); First Aid (5, 5): $\sqrt{(0.25 + 9)} \approx 3.04$ units ≈ 152 ft ✓
11	Recommend Landing or Boat Dock and justify with angle, ride time, cost of cable ($8 per ft), and safety.	Answers vary; Landing: ≈ 502 ft ($4,016), 5.5°; Dock: ≈ 453 ft ($3,624), 6.1°.

Reflection

#	Prompt	Look for
R1	Why does a longer horizontal distance make a smaller descent angle?	The opposite side stays 48 ft while the adjacent side grows, so tan θ decreases.
R2	How did special right triangles let you find exact answers?	Their side ratios (1 : 1 : √2 and 1 : √3 : 2) give exact lengths without a calculator.